

AI-Driven Automated Evidence Correlation in Cyber Forensics

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Abstract— The rapid growth of cybercrime and the increasing complexity of digital infrastructures have significantly challenged traditional digital forensic investigation methods. Modern cyber incidents generate massive volumes of heterogeneous evidence across endpoints, networks, cloud platforms, and user devices. Conventional forensic approaches rely heavily on manual artifact examination, rule-based correlation, and investigator-driven reasoning, which are time-consuming, error-prone, and difficult to scale in distributed environments. To address these limitations, recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Graph Neural Networks (GNNs) have introduced intelligent, data-driven techniques for automated evidence analysis and correlation. These techniques enable multimodal data fusion, relational reasoning through knowledge graphs, probabilistic link prediction, anomaly detection, and explainable decision support. This seminar presents a comprehensive review of AI-based digital forensic research and proposes an AI-Driven Automated Evidence Correlation framework that integrates multimodal feature extraction, Digital Forensic Knowledge Graph construction, GNN-based relational modeling, and explainable AI mechanisms. The proposed approach enhances investigation speed, correlation accuracy, scalability, and legal defensibility while supporting human-in-the-loop validation. The findings indicate that AI-driven forensic systems have strong potential to transform digital investigations into more intelligent, scalable, transparent, and reliable processes capable of addressing modern cyber threats.

Index Terms— Digital Forensics, Artificial Intelligence, Evidence Correlation, Graph Neural Networks, Knowledge Graphs, Multimodal Learning, Explainable AI, Cyber Investigation, DFIR.

I. INTRODUCTION

The rapid expansion of digital technologies, cloud computing, mobile platforms, and interconnected network infrastructures has significantly increased the volume and complexity of cybercrime. Modern cyber incidents generate vast amounts of heterogeneous digital evidence, including disk artifacts, registry entries, network traffic logs, cloud audit records, malware traces, and user activity logs. Digital forensics is the process of identifying, preserving, analyzing, and presenting such evidence to reconstruct cyber incidents and support legal proceedings. However, traditional forensic investigation methods rely heavily on manual artifact examination, rule-based correlation, and investigator-driven reasoning, which can be time-consuming, error-prone, and difficult to scale in large and distributed environments.

Efficient evidence correlation plays a critical role in digital investigations. Cyber attacks often involve

multi-stage operations such as initial intrusion, privilege escalation, lateral movement, data exfiltration, and persistence mechanisms. These activities leave traces across multiple platforms and devices. Manually linking artifacts from diverse sources to reconstruct a coherent attack timeline is a complex and resource-intensive task. Conventional forensic tools focus primarily on artifact extraction rather than intelligent cross-source reasoning, limiting their ability to uncover hidden relationships and causal attack chains.

The emergence of Artificial Intelligence (AI) and Machine Learning (ML) has introduced new possibilities for intelligent, data-driven forensic systems. AI-based approaches can analyze large-scale structured and unstructured datasets, identify hidden patterns, detect anomalies, and automatically correlate related entities across multiple evidence sources. Advanced techniques such as Deep Learning and Graph Neural Networks (GNNs) enable relational reasoning within knowledge

graphs, supporting automated link prediction and probabilistic evidence scoring. Unlike traditional rule-based methods, AI-driven systems can adapt to evolving attack patterns and improve performance through continuous learning.

Recent research demonstrates the effectiveness of machine learning models such as Random Forest, Support Vector Machines (SVM), and ensemble techniques in malware classification and anomaly detection, while deep learning architectures including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and Transformer models show strong capability in analyzing unstructured and sequential forensic data. Furthermore, knowledge graph modeling and explainable AI mechanisms enhance transparency, provenance tracking, and legal defensibility in AI-assisted investigations.

This review aims to analyze recent AI-based approaches for automated evidence correlation in digital forensics, categorize methodologies based on data modality and algorithmic techniques, compare strengths and limitations of existing models, identify research gaps in current forensic automation frameworks, and propose a comprehensive AI-Driven Automated Evidence Correlation architecture to improve investigation efficiency, scalability, transparency, and reliability in modern cyber environments.

II. RELATED WORKS

The application of Artificial Intelligence (AI) in cyber forensics—particularly in **automated evidence correlation**—has expanded significantly in recent years. Researchers have explored multiple forensic data modalities including disk artifacts, system and network logs, cloud audit trails, malware binaries, user activity traces, and unstructured communication data (emails, chats, documents) to improve investigation efficiency and reliability. Instead of relying solely on manual artifact examination and rule-based correlation, AI-based approaches aim to build **automated, objective, scalable, and explainable forensic reasoning systems** that can link related evidence across distributed digital environments.

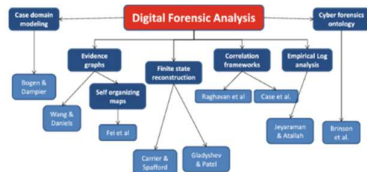


Fig.1. Taxonomy of AI techniques and forensic data modalities used for automated evidence correlation in cyber forensics

The overall categorization of AI techniques and forensic data modalities is illustrated in Fig. 1.

A. Automated Forensic Pipelines and Rule-Based Correlation Frameworks

Early research in automated digital investigations introduced structured forensic pipelines to support evidence extraction, normalization, and correlation across heterogeneous sources. Such frameworks focus on building systematic workflows where artifacts from different tools and targets can be unified into a common representation for correlation and reporting. These systems typically perform preprocessing tasks such as timeline extraction, metadata parsing, and evidence indexing before applying correlation rules to identify relationships among artifacts.

While rule-based approaches provide clarity and deterministic outputs, they are limited in handling uncertainty, evolving attack strategies, and complex multi-stage cyber incidents. Their effectiveness largely depends on predefined correlation rules and expert-defined patterns, which may not generalize well to novel or stealthy attack behaviours.

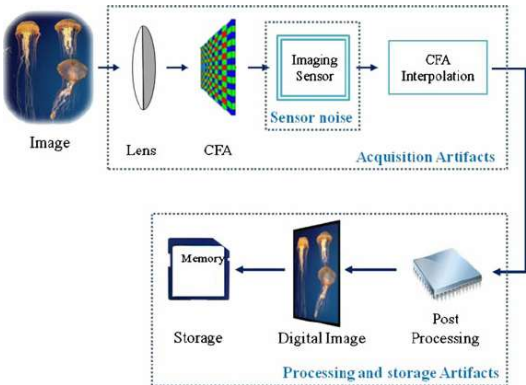


Fig. 2. Automated evidence processing pipeline showing acquisition, artifact extraction, indexing, and correlation stages.

Typical workflow stages in early automated forensic pipelines are shown in Fig. 2.

B. Graph-Based Evidence Modeling and Relationship Reasoning

Graph-based forensic approaches represent digital evidence as interconnected structures where entities (users, files, IP addresses, devices, processes, timestamps) are modeled as **nodes** and their interactions (executed, accessed, connected-to, downloaded-from, modified) are modeled as **edges**. This representation supports structured reasoning, timeline reconstruction, and visualization of complex relationships across evidence sources.

Graph modeling enables investigators to trace multi-hop connections, identify hidden associations, and reconstruct event sequences that are difficult to observe from raw logs alone. Graph traversal, link analysis, and temporal graph construction have been used to reconstruct attack narratives in network intrusions and distributed incidents. However, many graph-based methods still rely on deterministic reasoning and may struggle with noisy data, incomplete logs, or uncertain relationships.

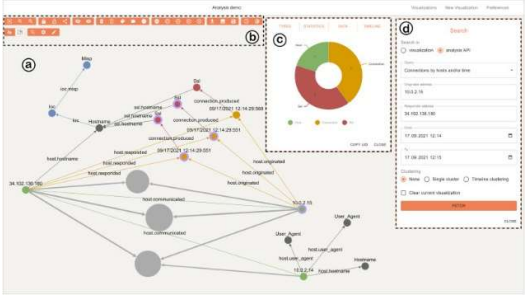


Fig. 3. Graph-based evidence representation linking users, devices, files, and network events.

The graph modeling of forensic evidence and relationship linking is illustrated in Fig. 3.

C. Digital Forensic Knowledge Graphs and Semantic Reasoning

Recent studies extend graph-based forensic models into **Digital Forensic Knowledge Graphs (DFKGs)** by incorporating semantic meaning through ontologies and structured relationship definitions. Knowledge graph approaches improve forensic reasoning by enabling semantic querying, hypothesis testing, and provenance-aware tracking of evidence lineage. This is particularly important for ensuring transparency, maintaining chain-of-custody, and supporting legally defensible investigation outcomes.

Knowledge graphs provide structured intelligence by connecting evidence artifacts with context and meaning, allowing investigators to query relationships such as “Which processes accessed this file before exfiltration?” or “Which device communicated with a suspicious IP during the attack window?” Despite these advantages, many DFKG frameworks emphasize representation and querying but provide limited automated inference unless integrated with machine learning-based correlation mechanisms.

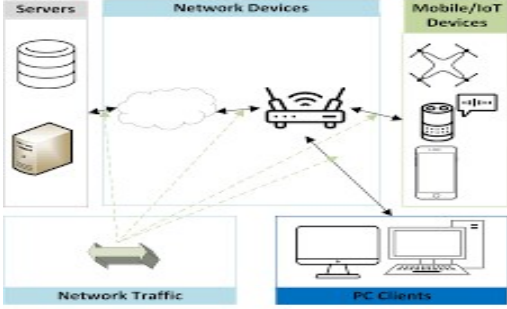


Fig. 4. Semantic evidence reasoning using a Digital Forensic Knowledge Graph (DFKG) with provenance tracking.

Semantic linking and provenance-aware modeling are illustrated in Fig. 4.

D. Machine Learning and Deep Learning for Forensic Intelligence

Machine Learning and Deep Learning have been applied extensively in digital forensics to improve automation in malware classification, anomaly detection, log analysis, and suspicious activity identification. Traditional ML models such as Support Vector Machines (SVM), Random Forest, and ensemble techniques perform well on structured forensic datasets such as tabular logs and extracted metadata. These models reduce manual triage by classifying artifacts into suspicious or benign categories and prioritizing evidence for investigators.

Deep learning architectures such as Convolutional Neural Networks (CNN) are widely used in malware analysis and binary classification, while Long Short-Term Memory (LSTM) networks and Transformer-based models are effective in analyzing sequential and textual forensic data such as event logs, system traces, emails, and chat records. Although these models often achieve high accuracy, their effectiveness in real-world cases depends on dataset diversity, robustness to adversarial manipulation, and integration into a complete correlation pipeline rather than operating as isolated classifiers.

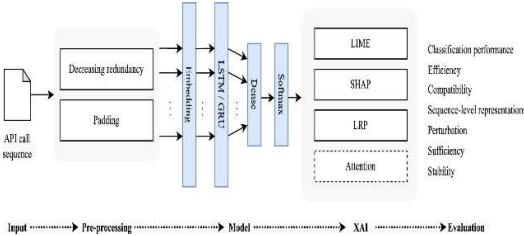


Fig. 4. AI-based forensic intelligence models applied to malware, log sequences, and behavioral traces.

Deep learning and ML-based forensic classification workflows are shown in Fig. 5.

III. COMPARATIVE ANALYSIS

The growing body of research on AI-based automation in cyber forensics demonstrates the increasing effectiveness of machine learning, graph-based modeling, and deep learning techniques for evidence analysis and correlation. However, performance and practical usefulness vary significantly depending on the type of forensic data, preprocessing strategies, graph construction quality, and model complexity. This section provides a comparative analysis of commonly used approaches in AI-driven automated evidence correlation.

A. Comparison of AI Architectures and Correlation Approaches

1) Rule-Based Automated Forensic Pipelines

Early forensic automation frameworks rely on structured pipelines for artifact ingestion, normalization, and deterministic correlation using predefined rules. These pipelines are effective for organizing evidence from multiple sources and generating consistent outputs. They work well when attack patterns are known and artifacts follow predictable structures.

Strengths: clear reasoning, deterministic outputs, supports structured processing and repeatability.
Limitations: rigid logic, poor adaptability to evolving attacks, limited ability to discover hidden relationships, and high dependence on expert-defined rules.

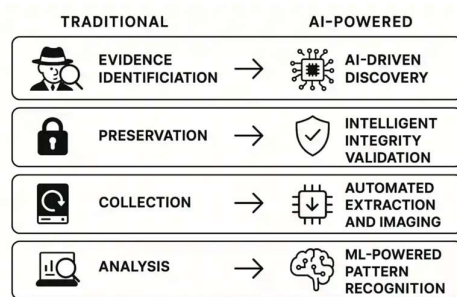


Fig. 5. Rule-based automated pipeline for artifact ingestion and evidence correlation.

A typical pipeline-based correlation workflow is depicted in Fig. 5.

2) Graph-Based Evidence Modeling and Link Analysis

Graph-based methods model evidence as nodes and relationships as edges, enabling investigators to visualize connections and reconstruct event sequences across devices, users, files, and network

entities. Graph traversal and link analysis provide structured reasoning and support multi-hop relationship discovery, which is difficult in raw logs.

Strengths: strong relationship visualization, effective for event reconstruction, supports temporal linking and narrative building.

Limitations: often deterministic, limited handling of uncertainty, scalability challenges for large graphs, and depends on accurate entity linking.

3) Digital Forensic Knowledge Graphs (DFKGs)

Knowledge graph approaches extend graph modeling by adding semantic meaning, ontology-based representation, and provenance tracking. DFKGs improve explainability by representing evidence in a structured form suitable for hypothesis testing and querying. They are particularly useful for producing traceable and legally defensible investigation outputs.

Strengths: semantic reasoning, structured query support, provenance preservation, improved explainability.

Limitations: limited automated inference without ML integration, knowledge engineering complexity, and dependence on well-defined ontologies.

4) Machine Learning-Based Forensic Classification Models

Machine learning algorithms are widely applied for malware detection, anomaly detection, log classification, and triage prioritization. Models such as Random Forest, SVM, Decision Trees, and ensemble methods perform well for structured forensic datasets extracted from logs, metadata, and behavior traces. These models help reduce investigator workload by classifying suspicious artifacts early.

Strengths: strong performance on structured datasets, faster triage, reduced manual analysis, scalable classification.

Limitations: often used as isolated classifiers, weak multimodal correlation capability, limited interpretability, and sensitivity to dataset imbalance.

5) Deep Learning and GNN-Based Correlation Models

Deep learning models such as CNN, LSTM, and Transformer architectures are effective in handling unstructured forensic data (malware binaries, sequential logs, communication data). Graph Neural Networks (GNNs) further enhance correlation by learning directly on evidence graphs and performing link prediction and relationship scoring. These

approaches improve automated discovery of hidden relationships across multi-source evidence.

Strengths: superior pattern extraction, effective for unstructured and sequential evidence, automated link prediction, supports probabilistic correlation scoring.

Limitations: high computational complexity, requires large labeled datasets, interpretability challenges, and potential vulnerability to adversarial manipulation or poisoned evidence.

B. Comparison of Forensic Data Modalities

Log and metadata-based systems are relatively scalable and widely available, but logs may be incomplete, noisy, or manipulated during attacks. Malware and binary evidence provides strong indicators of compromise, yet deep learning models require diverse and well-labeled malware datasets. Cloud forensic data is highly relevant for modern investigations but is fragmented across distributed services and may be ephemeral. Communication artifacts such as emails and chats provide contextual intelligence but require NLP-based extraction and may raise privacy concerns. Multimodal evidence integration offers the highest investigative value because it combines complementary forensic signals; however, it introduces challenges in data synchronization, normalization, and computational efficiency.

C. Performance Trends

Across studies, AI-based approaches generally outperform purely manual and rule-based forensic correlation methods in terms of scalability and relationship discovery. Graph-based and knowledge graph methods significantly strengthen structured reasoning and improve explainability, while machine learning models enhance classification performance and speed up evidence triage.

Deep learning architectures and multimodal frameworks show higher accuracy in detecting suspicious patterns and correlating heterogeneous evidence, especially when investigations involve both structured logs and unstructured data. In particular, GNN-based correlation models represent the strongest emerging trend because they learn relational structures directly and enable automated link prediction within forensic knowledge graphs.

Table 1: Comparison of AI Approaches for Automated Evidence Correlation in Cyber Forensics

AI Approach	Data Type	Algorithms Used	Strengths	Limitations	Accuracy / Performance
Rule-based pipelines	Normalized artifacts/logs	Rules / heuristics	Simple, explainable	Rigid, brittle	Baseline
Graph link analysis	Entities + relations	Graph traversal/linking	Good reconstruction	Scaling, entity errors	Moderate-Good
Forensic Knowledge Graphs	Semantic + provenance	Ontologies / queries	Explainable reasoning	Build/maintain heavy	Good (varies)
ML classifiers	Structured features	RF / SVM / DT	Fast triage	Weak cross-source links	Good for classification
DL + GNN models	Unstructured + graphs	CNN/LSTM/Tx + GNN	Strong link prediction	Data/compute heavy	High (trend)
Multimodal fusion	Multi-source evidence	Hybrid DL + GNN	Best overall correlation	Integration costly	Highest (trend)

As seen in Table 1, multimodal evidence fusion tends to perform best because it merges endpoint, network, cloud, and malware evidence into one view, improving correlation and revealing links that isolated tools can miss in complex cases. However, multimodal systems demand heavy integration, time alignment, and more compute. In contrast, single-stream methods—rule pipelines, ML classifiers, or deep models on one artifact type—are simpler to deploy and effective for targeted triage, but they often miss cross-source relationships needed for full reconstruction. Results vary across studies with different datasets and evaluation setups, and real investigations may suffer from missing, noisy, or tampered evidence.

IV. RESEARCH GAPS

Although Artificial Intelligence has shown promising results in automating digital forensic analysis and evidence correlation, several critical research gaps remain unaddressed. Identifying these limitations is essential for developing more reliable, scalable, explainable, and legally defensible AI-driven forensic systems capable of operating in real-world cyber investigations.

A. Limited Multimodal Evidence Integration

Many existing forensic AI studies focus on a single evidence modality such as network logs, malware binaries, or system events. While these approaches show strong task-specific performance, real cyber incidents are complex and distributed, producing

evidence across endpoints, cloud platforms, user devices, communications, and network infrastructure. Limited research has developed robust multimodal fusion frameworks that integrate disk artifacts, registry traces, cloud audit logs, user behaviour analytics, emails/chats, and threat intelligence into a unified correlation pipeline. The absence of comprehensive multimodal integration reduces investigation completeness and increases the likelihood of missing cross-source relationships required for accurate incident reconstruction.

B. Lack of Standardized and Realistic Forensic Datasets

A major limitation in current research is the reliance on small, controlled, or benchmark datasets that may not reflect real-world investigation conditions. Many studies use simulated attacks, limited malware sets, or incomplete log collections, which restricts generalizability. In practical investigations, evidence can be noisy, incomplete, inconsistently formatted, and distributed across multiple jurisdictions. In addition, labelled forensic datasets for correlation tasks (ground-truth relationships among artifacts) are scarce, making it difficult to train and evaluate advanced AI correlation engines such as GNNs. This gap highlights the need for ethically curated, continuously updated, large-scale datasets that reflect real operational forensic complexity.

C. Limited Explainability and Legal Defensibility

Most deep learning and graph learning models operate as “black boxes,” producing correlation scores or suspicious link predictions without clear reasoning. In cyber forensics, transparency is essential because findings must be defensible in legal proceedings. Many AI-based forensic tools still lack integrated Explainable AI (XAI) mechanisms that can provide traceable evidence paths, confidence justification, and human-readable reasoning aligned with forensic reporting standards. Without interpretability, investigators may hesitate to rely on AI-driven correlations, and courts may challenge the admissibility of AI-generated conclusions.

D. Weak Provenance and Chain-of-Custody Integration

Forensic investigations require strict preservation of evidence integrity, including chain-of-custody documentation, provenance tracking, and audit logging. While knowledge graph approaches sometimes represent provenance structurally, many AI correlation models do not integrate evidence lineage directly into inference and reporting

workflows. This creates a gap between AI automation and forensic legal requirements. A unified framework that combines automated correlation with tamper-evident provenance, integrity verification, and auditable transformation history remains underdeveloped.

E. Limited Temporal and Causal Reasoning

Correlation alone is not sufficient in complex cybercrime investigations; investigators must reconstruct timelines and causal chains to explain how an incident occurred. Many existing studies correlate artifacts based on similarity or relational structure but do not incorporate strong temporal validation, causal inference, or uncertainty-aware timeline reconstruction. Cyber incidents often involve multi-stage attack chains where events occur across different systems and time zones, and attackers may manipulate timestamps. Therefore, integrating temporal-causal reasoning into AI-driven evidence correlation remains a major research gap.

F. Adversarial Manipulation and Anti-Forensic Challenges

Cyber attackers frequently use anti-forensic techniques such as log wiping, obfuscation, timestamp manipulation, data injection, and misleading artifact creation to disrupt investigations. Many AI-based forensic models are not evaluated against adversarial conditions and may be vulnerable to poisoning attacks or manipulated evidence inputs. Robustness to adversarial evidence tampering, secure model training, and validation mechanisms are insufficiently explored in existing forensic AI research. Addressing this gap is critical for trustworthy real-world deployment.

G. Privacy, Ethics, and Regulatory Constraints

Digital forensic data often contains sensitive personal and organizational information, including user activities, communications, locations, and private files. AI-driven systems that process such evidence must comply with privacy laws, organizational policies, and ethical standards. However, limited research has explored privacy-preserving approaches such as federated learning, secure multi-party computation, anonymization techniques, or controlled access frameworks within forensic AI pipelines. Ensuring fairness, ethical governance, and compliance with regulations is essential for responsible deployment.

V. PROPOSED MULTIMODAL AI FRAMEWORK FOR AUTOMATED EVIDENCE CORRELATION

Based on the limitations identified in existing cyber forensic studies, this seminar proposes a comprehensive **multimodal Artificial Intelligence framework for automated evidence correlation in cyber forensics**. The proposed system aims to integrate heterogeneous forensic evidence sources, improve correlation accuracy and investigative reliability, support explainable and legally defensible reasoning, and ensure secure handling of sensitive evidence data. The framework is designed as an intelligent decision-support architecture that assists investigators in reconstructing cyber incidents efficiently across distributed digital environments.

A. Overview of the Proposed Framework

The proposed framework consists of six major components: **multisource evidence collection, preprocessing and normalization, multimodal feature extraction, knowledge graph construction, GNN-based correlation and reasoning, and an explainability plus provenance module with investigator dashboard support**. The architecture is intended to operate within cyber forensic and DFIR workflows where evidence is collected from multiple devices and platforms. The system produces correlated evidence links, reconstructed timelines, and transparent reasoning outputs suitable for investigative reporting and legal documentation.

B. Multimodal Evidence Collection

To capture the distributed and heterogeneous nature of cybercrime evidence, the framework integrates the following forensic data sources:

Digital endpoint evidence includes disk artifacts, file system metadata, registry entries, and memory traces collected from compromised systems. Network evidence includes packet captures, flow records, firewall logs, IDS alerts, and DNS logs that represent communication behaviour and potential intrusion paths. Cloud forensic evidence includes audit logs, API access records, virtual machine activity traces, and storage event logs from cloud service providers. Malware and executable evidence includes binaries, hashes, behavioural traces, sandbox reports, and persistence mechanisms used by attackers. Communication artifacts include emails, chat logs, documents, and web browsing

traces that may provide contextual and attribution evidence. User and access evidence includes authentication logs, privilege escalation events, session records, and user behaviour analytics that support insider threat and compromise investigation.

By combining these modalities, the system aims to capture both technical traces and contextual indicators of cyber incidents, improving investigative completeness and reducing evidence fragmentation across platforms.

C. Evidence Preprocessing and Feature Engineering

Each modality undergoes specialized preprocessing to convert raw evidence into structured, machine-readable representations. Disk and file artifacts undergo parsing, timeline extraction, metadata tagging, hashing, and deduplication. Network logs undergo session reconstruction, noise filtering, timestamp normalization, and feature extraction such as traffic frequency, protocol patterns, and connection anomalies. Cloud logs require API event normalization, identity mapping, and temporal alignment due to distributed storage and multiple log formats. Malware evidence undergoes static feature extraction (hashes, strings, imports) and dynamic behavioural extraction (system calls, network callbacks, file modifications). Textual artifacts such as emails and chats undergo Natural Language Processing steps including entity extraction, keyword tagging, and relationship detection.

Feature engineering techniques transform these processed artifacts into structured vectors, temporal sequences, and graph attributes suitable for AI-based modeling and correlation.

D. Hybrid AI Architecture for Correlation and Reasoning

The proposed system integrates multiple AI components inspired by successful approaches in forensic automation and multimodal intelligence. A hybrid architecture is adopted because different evidence types require different modeling strategies. The framework incorporates deep learning models for unstructured evidence (such as malware binaries, text logs, and multimedia artifacts) and machine learning models for structured forensic logs and metadata. Convolutional Neural Networks (CNNs) are used for malware binary classification and artifact pattern extraction, while Long Short-Term Memory (LSTM) models support sequential modeling for time-series logs and event flows. Transformer-based models and NLP pipelines assist

in extracting entities and intent from emails, chat messages, and unstructured reports.

To enable cross-source correlation, all extracted entities and features are mapped into a unified graph representation, and a **Graph Neural Network (GNN)** is applied to learn relational patterns directly on the evidence graph. The GNN performs link prediction, node classification, and relationship scoring to identify hidden evidence connections. This hybrid architecture supports both classification (detecting suspicious artifacts) and correlation (linking related artifacts across sources), enabling reconstruction of multi-stage cyber incidents.

E. Explainable AI (XAI) and Provenance Integration

To address the black-box limitation of deep learning and GNN-based systems, the framework integrates Explainable AI mechanisms within the correlation workflow. The explainability module generates human-readable reasoning outputs by highlighting key evidence nodes, relationship paths, correlation confidence scores, and supporting artifacts contributing to each conclusion. Graph-based explanations provide traceable “evidence chains” that show how related artifacts are connected across sources.

In addition, the framework embeds provenance-aware tracking mechanisms to preserve chain-of-custody and legal defensibility. Each evidence artifact is tagged with acquisition metadata, hashing integrity records, transformation history, and audit logs. This ensures that automated correlations remain traceable and admissible, supporting forensic reporting standards and courtroom requirements.

F. Secure and Privacy-Preserving Deployment

Forensic evidence often contains sensitive organizational and personal data. Therefore, the proposed framework emphasizes secure handling and controlled access mechanisms. Evidence storage and processing are designed with strict logging, role-based access control, and tamper-evident integrity verification. For cross-institution collaboration (such as large-scale threat intelligence sharing), privacy-preserving learning strategies such as federated learning can be adopted, where model improvements are shared without transferring raw evidence outside the organization. This approach helps maintain confidentiality while enabling continuous model retraining and adaptation to evolving cyber threats.

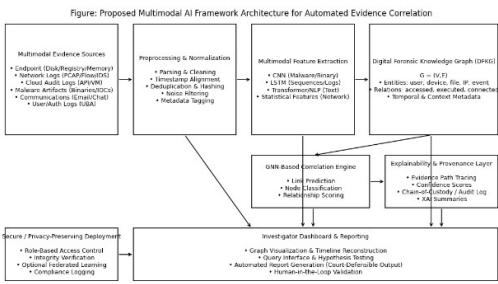


Fig: Proposed Multimodal AI Framework Architecture

G. Expected Advantages

The proposed AI-driven automated evidence correlation framework offers several advantages over traditional manual and single-source forensic approaches. By integrating heterogeneous evidence sources such as disk artifacts, network logs, cloud audit trails, malware traces, communications, and user activity records, the system improves correlation accuracy through complementary evidence fusion. The knowledge graph representation enables structured semantic modeling of entities and relationships, supporting clearer event reconstruction and hypothesis testing across distributed environments. The integration of Graph Neural Network–based correlation strengthens relationship discovery by enabling probabilistic link prediction and hidden pattern detection that rule-based methods may overlook. In addition, the explainability and provenance components improve transparency by providing traceable evidence paths, confidence scoring, and chain-of-custody preservation, which are essential for investigator trust and legal defensibility. The human-in-the-loop dashboard design ensures that AI outputs support forensic analysts rather than replace them, improving accountability and reducing the risk of incorrect automated conclusions. Furthermore, the scalable architecture supports large-volume investigations in enterprise and cloud environments, making the framework practically deployable for modern cyber forensic and DFIR operations.

VI. CONCLUSION AND FUTURE WORK

Automated evidence correlation has become essential in modern cyber forensics due to the increasing complexity of cybercrime and the massive volume of heterogeneous digital artifacts generated across endpoints, networks, and cloud platforms. Traditional forensic approaches, although

reliable for structured cases, often rely on manual analysis and predefined correlation rules, which reduces efficiency when investigating distributed and multi-stage incidents.

This seminar reviewed major advancements in AI-driven forensic automation, including rule-based forensic pipelines, graph-based evidence modeling, Digital Forensic Knowledge Graphs for semantic reasoning, and machine/deep learning methods for classification and anomaly detection. The analysis shows that graph and knowledge-graph methods strengthen structured reasoning and interpretability, while deep learning improves pattern extraction from unstructured forensic data.

However, important challenges remain, such as limited availability of standardized real-world forensic datasets, complexity in multimodal evidence fusion, high computational cost of graph learning models, adversarial manipulation risks, and insufficient temporal-causal reasoning. In addition, forensic investigations require transparent and legally defensible outputs, making explainability and provenance tracking critical.

To address these limitations, the proposed framework integrates multimodal evidence ingestion, preprocessing and feature extraction, knowledge graph construction, GNN-based correlation, explainable reasoning, provenance-aware chain-of-custody support, and investigator-centered decision making. Future work should focus on building realistic datasets, improving robustness and temporal reasoning, enhancing forensic-aligned explainability, and validating deployment in real operational environments under privacy and regulatory constraints.

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